

WTW 158 Calculus
ASSIGNMENT: Exam July 2021

When submitting this assignment, you agree to the following:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

This assignment consists of 2 parts.

Section B: 15 marks (minimum 30 minutes)

Section C: 15 marks (35 minutes)

INSTRUCTIONS:

1. The exam paper is an assignment consisting of 2 parts.
2. Draw a line with a ruler at the end of each question.
3. Write your surname, initials and student number clearly in the beginning of your assignment.
4. **Show all your steps and calculations.**
5. Only handwritten assignments submitted in one PDF file will be accepted.
6. Before submitting, make sure that you are satisfied with your answers and that you submit the correct document.
7. No late assignments will be accepted.
8. No assignments sent via e-mail will be accepted.
9. Calculators **may not** be used.

Section C: 35 minutes**Question 4** [5 marks]

Function f consists of the bottom half of the circle with center the origin and radius 2 and the line segment from the point $(2, 0)$ to the point $(4, 5)$. Explain all your steps when you do the following:

(4.1) Find an expression for the function f . (2)

(4.2) Find the total area between the graph of the function f and the x -axis. (2)

(4.3) Give the value of (1)

$$\int_{-2}^4 f(x) dx.$$

Question 5 [6 marks]

Consider function $f(x) = x^3(x - 4)$. Motivate all your answers.

(5.1) Use the Closed Interval Method to find the absolute extreme values of f on $[-2, 2]$. (3)

(5.2) Determine if f has any local extreme values or inflection points on $[-2, 2]$. (3)

Question 6 [4 marks]

A wooden container with a square base and an open top must have a volume of $32\,000\text{ cm}^3$. Determine the dimensions of the box that minimize the amount of paint needed to paint the outside areas of the container.